

Logistic Regression in R (Ebay Auction)

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Context

Let's take a look at the data located in `ebayAuctions.csv`. Each row represents an auction, with the columns specifying various auction characteristics, including information about the seller (`SellerRating`) and the item (`Category`). Our goal here is to predict whether the auction will turn out to be competitive (`Competitive`), here defined as receiving 2 or more bids. This type of prediction is crucial for any type of automated trading algorithm, as we would like to find auctions with relatively little competition in the hopes of getting a better final price.

Importing Data and Data Management

1. We'll be trying predict competitiveness using a logistic regression model. To get ready for modeling the following data management steps need to be completed .
- Since we're thinking about making our predictions of competitiveness at the beginning of the auction, we can really in good faith use `ClosePrice`; remove it from the data frame.
 - Convert the target to a logical.
 - Convert other variables to factors as necessary.
 - Partition the data using a 60-40 training-test split using random seed 1234.

```
df <- read.csv("ebayAuctions.csv")
df$Competitive <- as.logical(df$Competitive)
df$Category <- as.factor(df$Category)
df$Currency <- as.factor(df$Currency)
df$EndDay <- as.factor(df$EndDay)
df$ClosePrice <- NULL
```

Partition

2. setting up training partition as 60% of the data, and testing data as the rest.

```
set.seed(1234)
N <- nrow(df)
trainingSize <- round(N*0.6)
trainingCases <- sample(N, trainingSize)
training <- df[trainingCases,]
test <- df[-trainingCases,]
```

Building Model

3. Constructing a logistic regression model for Competitive as a function of all other available variables. Afterwards, also creating a new model by conducting backward stepwise variable elimination. We'll be using this smaller and more efficient model for the remainder of the questions here.

```
model1 <- glm(Competitive ~ ., data=training, family=binomial)
summary(model1)
```

Call:

```
glm(formula = Competitive ~ ., family = binomial, data = training)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	8.817e-01	4.471e-01	1.972	0.04860 *
CategoryAutomotive	-6.592e-01	3.172e-01	-2.078	0.03771 *
CategoryBooks	-2.760e-01	4.228e-01	-0.653	0.51383
CategoryBusiness/Industrial	2.526e-01	6.271e-01	0.403	0.68717
CategoryClothing/Accessories	-1.962e-01	3.611e-01	-0.543	0.58681
CategoryCoins/Stamps	-1.385e+00	5.344e-01	-2.592	0.00955 **
CategoryCollectibles	2.404e-01	2.795e-01	0.860	0.38968
CategoryComputer	3.260e-01	5.138e-01	0.635	0.52575
CategoryElectronics	1.269e+00	5.243e-01	2.421	0.01549 *
CategoryEverythingElse	-2.739e+00	1.122e+00	-2.441	0.01464 *
CategoryHealth/Beauty	-2.014e+00	5.056e-01	-3.982	6.83e-05 ***
CategoryHome/Garden	1.474e-01	3.450e-01	0.427	0.66924
CategoryJewelry	-5.448e-01	3.863e-01	-1.410	0.15851
CategoryMusic/Movie/Game	-1.846e-01	2.584e-01	-0.714	0.47495
CategoryPhotography	1.424e+01	2.989e+02	0.048	0.96200
CategoryPottery/Glass	-1.955e-01	6.629e-01	-0.295	0.76802
CategorySportingGoods	7.928e-01	3.911e-01	2.027	0.04266 *
CategoryToys/Hobbies	3.938e-01	2.895e-01	1.360	0.17367
CurrencyGBP	1.444e+00	5.096e-01	2.834	0.00459 **
CurrencyUS	-1.289e-01	2.095e-01	-0.615	0.53833
SellerRating	-5.198e-05	1.398e-05	-3.719	0.00020 ***
Duration	-9.979e-02	4.017e-02	-2.484	0.01298 *
EndDayMon	1.156e+00	2.192e-01	5.274	1.34e-07 ***
EndDaySat	-3.330e-01	2.361e-01	-1.411	0.15832
EndDaySun	2.208e-03	2.202e-01	0.010	0.99200
EndDayThu	-3.979e-01	4.634e-01	-0.859	0.39055
EndDayTue	4.677e-01	2.659e-01	1.759	0.07852 .
EndDayWed	3.060e-01	3.630e-01	0.843	0.39926
OpenPrice	-3.275e-03	2.768e-03	-1.183	0.23673

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 1627.2 on 1182 degrees of freedom
 Residual deviance: 1449.7 on 1154 degrees of freedom
 AIC: 1507.7

Number of Fisher Scoring iterations: 13

```
model2 <- step(model1)
```

Start: AIC=1507.74

Competitive ~ Category + Currency + SellerRating + Duration +
 EndDay + OpenPrice

	Df	Deviance	AIC
- OpenPrice	1	1451.6	1507.6
<none>		1449.7	1507.7
- Duration	1	1456.0	1512.0
- Currency	2	1458.8	1512.8
- SellerRating	1	1464.2	1520.2
- EndDay	6	1514.3	1560.3
- Category	17	1539.7	1563.7

Step: AIC=1507.62

Competitive ~ Category + Currency + SellerRating + Duration +
 EndDay

	Df	Deviance	AIC
<none>		1451.6	1507.6
- Duration	1	1458.3	1512.3
- Currency	2	1460.8	1512.8
- SellerRating	1	1465.5	1519.5
- EndDay	6	1516.8	1560.8
- Category	17	1545.8	1567.8

```
summary(model2)
```

Call:

```
glm(formula = Competitive ~ Category + Currency + SellerRating +  

  Duration + EndDay, family = binomial, data = training)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	9.098e-01	4.473e-01	2.034	0.041955 *
CategoryAutomotive	-8.060e-01	2.972e-01	-2.712	0.006696 **
CategoryBooks	-3.024e-01	4.231e-01	-0.715	0.474879
CategoryBusiness/Industrial	2.004e-01	6.266e-01	0.320	0.749106
CategoryClothing/Accessories	-2.380e-01	3.602e-01	-0.661	0.508753
CategoryCoins/Stamps	-1.432e+00	5.304e-01	-2.700	0.006940 **
CategoryCollectibles	2.274e-01	2.797e-01	0.813	0.416131

```

CategoryComputer      2.882e-01  5.123e-01   0.563 0.573734
CategoryElectronics   1.194e+00  5.207e-01   2.293 0.021865 *
CategoryEverythingElse -2.753e+00  1.122e+00  -2.454 0.014137 *
CategoryHealth/Beauty -2.027e+00  5.062e-01  -4.004 6.22e-05 ***
CategoryHome/Garden    1.267e-01  3.451e-01   0.367 0.713448
CategoryJewelry        -5.761e-01  3.858e-01  -1.493 0.135377
CategoryMusic/Movie/Game -1.968e-01  2.585e-01  -0.761 0.446453
CategoryPhotography    1.419e+01  3.000e+02   0.047 0.962270
CategoryPottery/Glass  -3.349e-01  6.452e-01  -0.519 0.603700
CategorySportingGoods   7.652e-01  3.905e-01   1.959 0.050062 .
CategoryToys/Hobbies    3.690e-01  2.891e-01   1.276 0.201883
CurrencyGBP            1.429e+00  5.103e-01   2.801 0.005093 **
CurrencyUS            -1.700e-01  2.072e-01  -0.820 0.411965
SellerRating          -5.082e-05  1.395e-05  -3.644 0.000268 ***
Duration              -1.035e-01  4.011e-02  -2.580 0.009888 **
EndDayMon              1.174e+00  2.188e-01   5.368 7.97e-08 ***
EndDaySat             -3.126e-01  2.350e-01  -1.330 0.183592
EndDaySun              9.965e-03  2.199e-01   0.045 0.963854
EndDayThu             -3.791e-01  4.640e-01  -0.817 0.413899
EndDayTue              4.828e-01  2.655e-01   1.819 0.068939 .
EndDayWed              3.222e-01  3.631e-01   0.887 0.374821
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

(Dispersion parameter for binomial family taken to be 1)

```

Null deviance: 1627.2  on 1182  degrees of freedom
Residual deviance: 1451.6  on 1155  degrees of freedom
AIC: 1507.6

```

Number of Fisher Scoring iterations: 13

```
options(scipen=100)
```

Interpreting

4. Interpret the coefficient for Duration.

Answer: When one unit of the Duration increases, the odds of the outcome of competitiveness decreases with the multiplicative of $e^{0.1}$.

5. Interpreting the coefficient for CurrencyGBP. Note that this variable is a dummy for Currency being British Pounds.

Answer: When we will use CurrencyGBP, the odds of the outcome of competitiveness will increase with the multiplicative of $e^{1.42931060}$, as compared to the use of base currency which is Euro.

Evaluating Model

6. Using a cut-off probability of 0.5, what is the error rate associated with your model? Use R to calculate it.

Answer: 0.36248

```
test$predictions <- predict(model2, test, type="response")
predictions <- test$predictions

test$predictionsTF <- (test$predictions >= 0.5)
predictionsTF <- test$predictionsTF

# store observed values too
observations <- test$Competitive
```

Error Rate

7. Showing the confusion matrix and then calculate the error rate manually by using R as a calculator using the numbers shown in the confusion matrix. The error rate is the proportion of incorrect predictions out of the total number of predictions:

Errorrate

$\text{Incorrect predictions} / \text{Total predictions} = 286\,789 \approx 0.3625$ The error is almost same.

```
table(predictionsTF, test$Competitive)
```

```
predictionsTF FALSE TRUE
FALSE      202  112
TRUE       174  301
```

```
error_rate <- sum(predictionsTF != observations)/nrow(test)
error_rate
```

```
[1] 0.3624842
```

Benchmark Error Rate

8. What is the Benchmark Error Rate? Does this show to be a useful model?

Answer: This is definitely a useful model as the error from the model is less than the benchmark error, thus the model does add value to our prediction tasks.

```
source("BabsonAnalytics.R")
error_bench <- benchmarkErrorRate(training$Competitive, test$Competitive)
error_bench
```

```
[1] 0.4765526
```

Sensitivity and Specificity

9. Calculating the sensitivity and specificity for the default cutoff probability (which is 0.5).

Answer:

```
sensitivity <- sum(predictionsTF == TRUE & observations == TRUE)/sum(observations == TRUE)
sensitivity
```

```
[1] 0.7288136
```

```
specificity <- sum(predictionsTF == FALSE & observations == FALSE)/sum(observations == FALSE)
specificity
```

```
[1] 0.537234
```

ROC Chart

10. Imagining that we now increase the cut-off probability from 0.5 to 0.6. Does the specificity increase or decrease? How do you know? When we increase the cut-off probability from 0.5 to 0.6, specificity generally increases, because when as the cut-off probability increases from 0.5 to 0.6, the model becomes more conservative in predicting the positive class (i.e., predicting "1"). This means the model will predict "1" (positive class) only if the probability is higher. As a result:

Fewer observations will be classified as positive and this leads to fewer false positives, which increases specificity.

11. Using the ROC Chart, does it show that the model is more or less useful than the benchmark? How do we know? Since the AUC = 0.68 and the observed values are a bit far above from the benchmark we can say that our model is fairly useful in predicting competitiveness but it can be surely said that the model can be improved to achieve even a better result by utilizing perhaps more variables or utilizing variables which have a higher significance in predicting the target variable.

```
source("BabsonAnalytics.R")
ROCChart(observations, predictions)
```

Warning: `qplot()` was deprecated in ggplot2 3.4.0.

